

EXPERIMENT 8

AIM:

To perform a **two-tailed Z-test** on the salary dataset to determine whether the average salary differs significantly from the specified population mean.

CODE:

```
import pandas as pd
```

```
import numpy as np
```

```
import scipy.stats as stats
```

```
data = pd.read_excel("Salary_data.xlsx")
```

```
sample_data = data["Salary"].values
```

```
population_mean = 70000
```

```
sample_mean = np.mean(sample_data)
```

```
sample_std = np.std(sample_data, ddof=1)
```

```
n = len(sample_data)
```

```
z_statistic = (sample_mean - population_mean) / (sample_std / np.sqrt(n))
```

```
p_value = 2 * (1 - stats.norm.cdf(abs(z_statistic)))
```

```
print(f"Sample Mean: {sample_mean:.2f}")
```

```
print(f"Z-Statistic: {z_statistic:.4f}")
```

```
print(f"P-Value: {p_value:.4f}")
```

```
alpha = 0.05
```

```
if p_value < alpha:
```

```
    print("Reject the null hypothesis: The average salary is significantly different  
from ₹70,000.")
```

```
else:
```

```
    print("Fail to reject the null hypothesis: There is no significant difference in  
average salary from ₹70,000.")
```

OUTPUT:

Sample Mean: 76003.00

Z-Statistic: 1.1994

P-Value: 0.2304

Fail to reject the null hypothesis: There is no significant difference in average salary from ₹70,000.

RESULT:

The Z-test was successfully performed, and the result showed no significant difference between the average salary and the specified population mean.